

PROJECT DOCUMENTATION

SmartHire GenAI

Resume Matching & AI Career Mentor

AI-Powered Resume Analysis, Semantic Job Matching, CV Improvement
and Retrieval-Augmented Career Guidance

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ABSTRACT

SmartHire GenAI is an AI-powered career assistance system that analyzes resumes, finds relevant job opportunities using semantic search, generates CV improvement suggestions, and provides grounded career guidance through an AI Career Mentor. The system accepts resumes in PDF and DOCX formats, extracts their text, and uses a Large Language Model (LLM) to create a structured profile containing name, skills, experience, education, and target role.

A pre-collected job dataset is embedded and indexed using FAISS so that candidate profiles can be matched with jobs semantically. The CV Improvement Generator compares a resume with a selected job and identifies missing skills, weak resume points, improved bullets, an improved summary, and actionable recommendations.

The AI Career Mentor uses Retrieval-Augmented Generation (RAG) with a career knowledge base to provide grounded career guidance. Guardrails validate inputs, reject unsafe or unrelated questions, and keep the mentor within its career-development scope. The project combines LLMs, embeddings, FAISS, LangChain, prompt engineering, document processing, RAG, and evaluation into one career-support system.

Keywords: Generative AI, Resume Analysis, Semantic Search, FAISS, Embeddings, Retrieval-Augmented Generation, LangChain, LLM, Career Mentor, Prompt Engineering, Guardrails, CV Improvement.

1. INTRODUCTION

Finding suitable employment can be difficult when candidates must compare their skills with many job descriptions. Traditional keyword-based approaches may miss relevant opportunities when similar concepts are expressed using different terminology. Candidates also need help identifying skill gaps, improving resume content, and choosing suitable career paths.

SmartHire GenAI integrates resume understanding, semantic retrieval, generative AI, and RAG. It creates a structured candidate profile, performs semantic job matching, generates targeted CV feedback, and provides grounded career guidance.

1.1 Problem Statement

A single intelligent system is needed to understand a candidate resume, identify relevant jobs, provide targeted CV improvement suggestions, and answer career questions using reliable supporting information.

1.2 Objectives

- Extract information from PDF and DOCX resumes.
- Create a validated structured JSON candidate profile.
- Perform semantic job matching using embeddings and FAISS.
- Generate targeted CV improvement suggestions without inventing candidate information.
- Build a RAG-based AI Career Mentor.
- Apply input validation, safety checks, and career-scope guardrails.
- Evaluate retrieval relevance and mentor answer quality.
- Maintain a reproducible GitHub project.

2. EXISTING AND PROPOSED SYSTEM

2.1 Existing System

Conventional career systems often rely on keyword matching, static filtering, or manual resume review. They may produce limited matches when resumes and job descriptions use different terminology and generally require separate tools for resume feedback and career guidance.

2.2 Proposed System

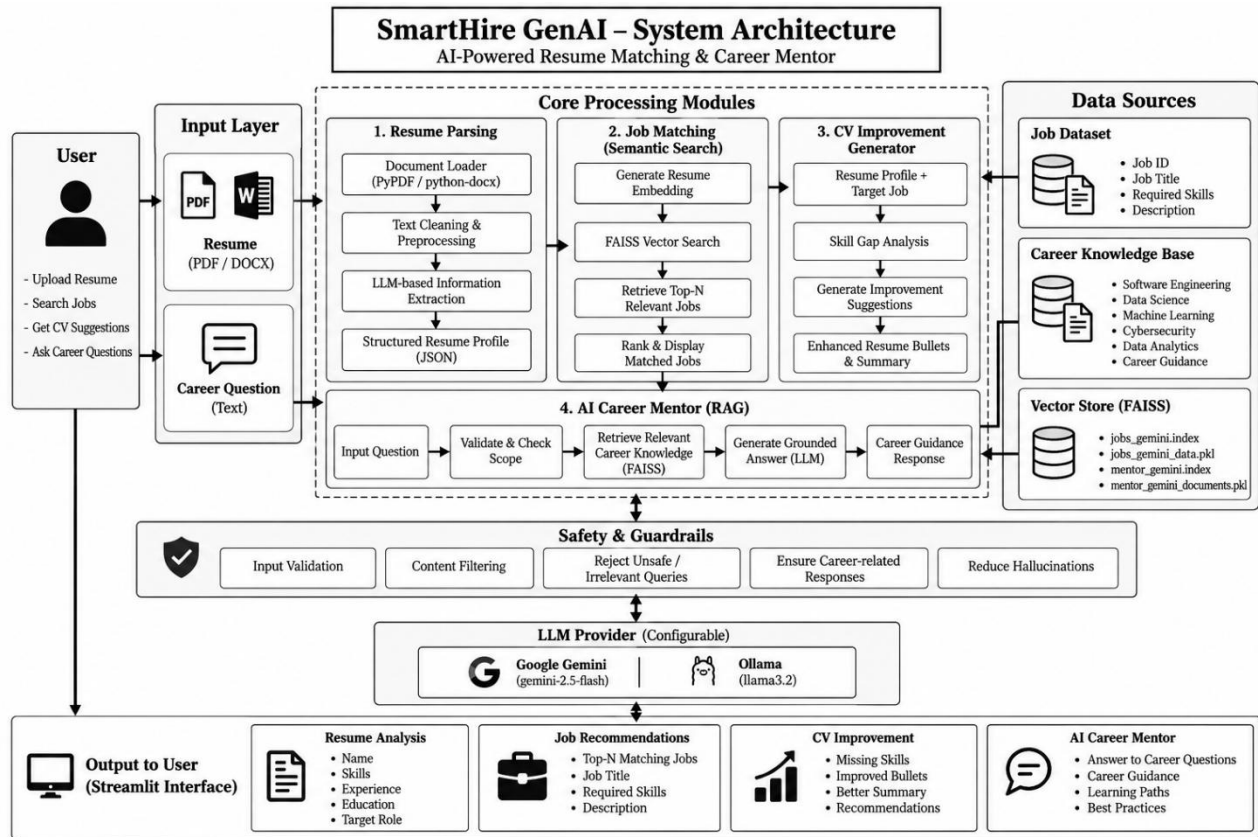
SmartHire GenAI integrates resume parsing, semantic job retrieval, CV improvement, and grounded career mentoring. It uses an LLM for structured extraction and generation, embeddings for semantic

representation, FAISS for vector retrieval, LangChain for AI workflows, and guardrails for controlled interaction.

3. SYSTEM REQUIREMENTS

Category	Requirement
Language	Python
AI Provider	Gemini for API/cloud use; Ollama support for local execution
Framework	LangChain
Vector Search	FAISS
Resume Processing	PyPDF and python-docx
Data Processing	Pandas and NumPy
Interface	Streamlit implementation included in repository
Configuration	.env / Streamlit Secrets
Input	PDF and DOCX resumes
Version Control	Git and GitHub

4. SYSTEM ARCHITECTURE



4.1 Major Components

Component	Function
Resume Loader	Reads PDF/DOCX files and extracts readable text.
Text Chunker	Splits longer content into manageable chunks.
Resume Parser	Creates and validates a strict JSON candidate profile.
Embedding Module	Creates vector representations of jobs, profiles and knowledge documents.
FAISS Job Index	Supports semantic job retrieval.
CV Generator	Produces job-specific resume improvement recommendations.
Mentor RAG Chain	Retrieves career documents before generating answers.
Guardrails	Validates inputs and limits mentor scope.
Evaluation	Measures retrieval and mentor response quality.

5. MODULE DESCRIPTION

5.1 Resume Parser

The loader extracts PDF text using PyPDF and DOCX text using python-docx. Text is cleaned and chunked. The LLM is instructed to return only JSON with name, skills, experience, education, and target_role. The response is parsed and validated for required fields and data types.

5.2 Semantic Job Search

The job dataset contains job_id, title, skills, and description. Job descriptions are embedded and stored in FAISS. A candidate profile is embedded during search and compared with stored vectors to return ranked jobs based on semantic similarity.

5.3 CV Improvement Generator

The resume and target job are passed to a reusable prompt. The LLM identifies missing skills, weak points, improved bullets, a professional summary, and actionable recommendations. The prompt prohibits invented qualifications, experience, certifications, or achievements.

5.4 AI Career Mentor — RAG

Career notes are chunked, embedded, and stored in a FAISS index. A user question is embedded, relevant documents are retrieved, and the retrieved context is supplied to the LLM. This keeps responses grounded in the project knowledge base.

5.5 Guardrails

Input validation, safety checks, and career-scope checks reject empty, oversized, unsafe, or unrelated questions. The mentor is restricted to career, resume, jobs, skills, interviews, education, and professional-development topics.

5.6 Evaluation

The evaluation module checks job retrieval relevance and assesses mentor responses for correctness, grounding, and helpfulness. It also checks guardrail behavior and unsupported-answer handling. Results are stored in reports/answer_quality.md.

6. SYSTEM WORKFLOW

Stage	Activity
1	Upload a PDF or DOCX resume.

Stage	Activity
2	Extract and clean resume text.
3	Send resume text to the LLM using a strict JSON prompt.
4	Validate the structured candidate profile.
5	Generate a candidate embedding.
6	Search the FAISS job index and return top-N jobs.
7	Select a target job for CV improvement.
8	Generate targeted CV recommendations.
9	Ask the Career Mentor a question.
10	Apply validation, safety, and scope checks.
11	Retrieve relevant career notes and generate a grounded answer.
12	Run evaluation and review answer quality.

7. DATASET AND KNOWLEDGE BASE

The system uses a pre-collected job dataset rather than live LinkedIn or Naukri scraping. The dataset is stored in `data/jobs/jobs_dataset.csv` and contains 100 records with job IDs, titles, skills, and descriptions.

Five sample resumes are included in PDF and DOCX formats for testing. The career knowledge base contains notes for Software Engineering, Data Analytics, Data Science, Machine Learning Engineering, and Cybersecurity.

8. AI AND PROMPT ENGINEERING

Prompt engineering is used for structured resume extraction, CV improvement, and the RAG mentor. Strict output requirements are used for JSON parsing. CV prompts require specific sections and prohibit fabrication. Mentor prompts use retrieved documents as context to support grounded answers.

8.1 Embeddings and FAISS

Embeddings convert text into numerical vectors representing semantic meaning. SmartHire uses embeddings for job descriptions, candidate profiles, and career documents. FAISS provides efficient vector similarity search. Gemini embeddings are supported for API use, while nomic-embed-text through Ollama is supported for local execution.

9. IMPLEMENTATION DETAILS

File / Folder	Purpose
src/ai_provider.py	Selects Gemini or Ollama and creates the LLM.
src/parsing/loader.py	Loads and chunks PDF/DOCX content.
src/parsing/resume_parser.py	Parses resume text into validated JSON.
src/search/embed.py	Provides the selected embedding model.
src/search/job_search.py	Loads/builds FAISS job indexes and searches jobs.
src/search/build_cloud_index.py	Builds Gemini-compatible FAISS indexes.
src/generate/prompts.py	Stores CV improvement prompts.
src/generate/cv_suggestions.py	Generates job-specific CV suggestions.
src/mentor/rag_chain.py	Runs the RAG career mentor.
src/safety/guardrails.py	Validates and scopes mentor questions.
src/evaluate.py	Runs evaluation and creates the answer-quality report.
app/streamlit_app.py	Contains the application interface.
data/	Contains jobs, resumes, and career notes.
vectorstore/	Stores FAISS indexes and associated data.
reports/	Stores evaluation/report outputs.

10. PROJECT STRUCTURE

```
smarthire-genai/  
├─ app/streamlit_app.py  
├─ data/  
│   ├─ jobs/  
│   ├─ resumes/  
│   └─ career_notes/  
├─ src/  
│   ├─ ai_provider.py  
│   ├─ parsing/  
│   ├─ search/  
│   └─ generate/
```

```
| ├── mentor/
| ├── safety/
| └── evaluate.py
├── vectorstore/
├── notebooks/
├── reports/
├── requirements.txt
├── .env.example
├── .gitignore
└── README.md
```

11. TESTING AND EVALUATION

Testing uses the sample resumes and representative career questions to verify each module and the mentor's scope.

Test Area	Expected Result
PDF Resume	Text is extracted and parsed into the required fields.
DOCX Resume	Text is extracted and parsed into the required fields.
Job Retrieval	Relevant jobs appear near the top of ranked results.
CV Improvement	Suggestions are job-specific and factual.
Career Question	Answer is grounded in retrieved career notes.
Out-of-scope Question	Question is rejected with a scope message.
Unsafe Question	Request is rejected by the safety guardrail.
Missing Knowledge	Mentor avoids fabricating unsupported information.

11.1 Evaluation Metrics

- Retrieval relevance / hit rate
- Mentor correctness
- Mentor grounding
- Mentor helpfulness
- Guardrail pass/fail behavior
- Unsupported-answer / hallucination checks
- Prompt quality before and after optimization

12. RESULTS AND DISCUSSION

The implemented system provides an end-to-end workflow for resume analysis, semantic job matching, CV improvement, and career mentoring. Resume parsing produces a structured profile, FAISS supports semantic job retrieval, the CV generator provides targeted guidance, and the RAG mentor uses career notes as a grounding source. Guardrails provide an additional control layer.

The project includes an evaluation implementation and answer-quality report. Actual numerical values should be taken directly from reports/answer_quality.md rather than being assumed in this document.

13. SECURITY AND RESPONSIBLE AI

- API keys are stored through environment variables or Streamlit Secrets rather than source code.
- The .gitignore file excludes sensitive configuration such as .env and Streamlit secrets.
- Mentor questions pass through validation and safety checks.
- Career scope is explicitly restricted to professional-development topics.
- The CV generator is instructed not to fabricate qualifications or achievements.
- RAG grounding is used to reduce unsupported mentor responses.

14. LIMITATIONS

- Job recommendations depend on the quality and coverage of the pre-collected dataset.
- The system does not obtain live job listings from LinkedIn or Naukri.
- LLM output quality can vary with model and API availability.
- Mentor answers are limited by the coverage of the career knowledge base.
- Scanned/image-only resumes may not yield readable text with the current loader.

15. FUTURE ENHANCEMENTS

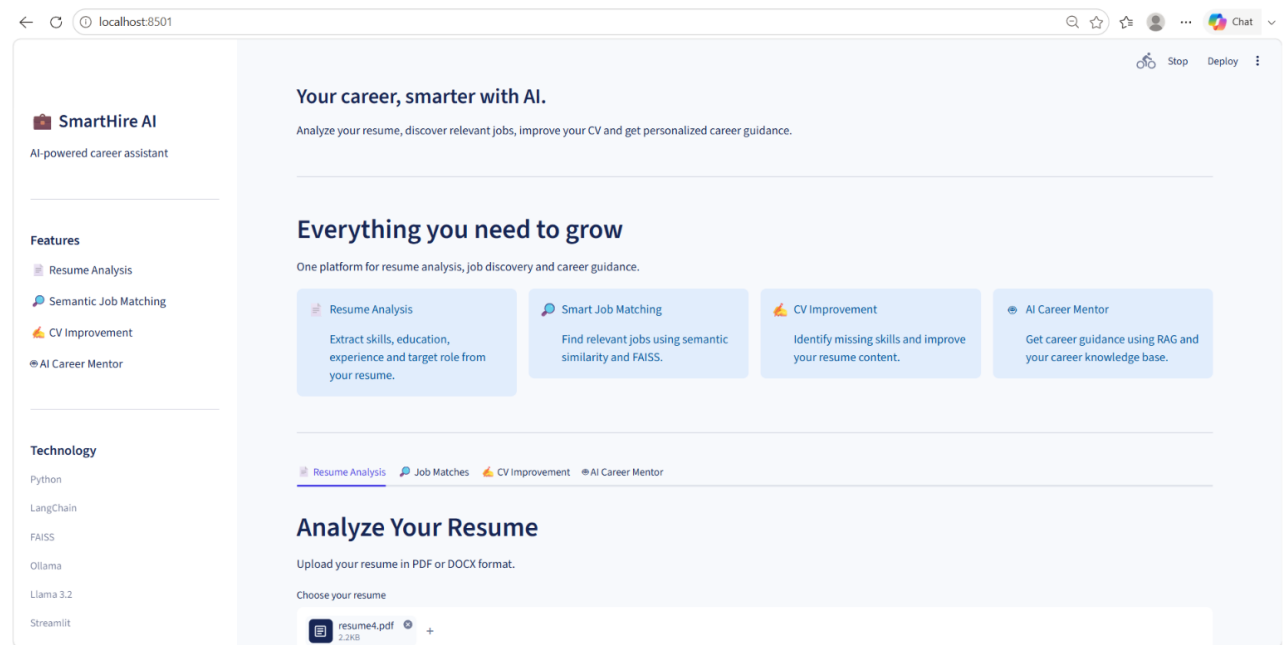
- Explainable reasons for individual job recommendations.
- Resume-to-job match percentage and skill-gap visualization.
- Citations to retrieved mentor sources.
- Personalized learning-path recommendations.
- Conversation memory for the mentor.
- Comparison of multiple embedding models.

- Larger job and career knowledge datasets.
- Improved factual resume rewriting.

16. CONCLUSION

SmartHire GenAI demonstrates how Generative AI, semantic search, vector databases, and Retrieval-Augmented Generation can be combined to build a practical career assistance system. The project goes beyond simple keyword matching by representing resumes and job descriptions semantically, while the CV improvement module provides targeted guidance and the RAG Career Mentor provides knowledge-grounded support. Validation, guardrails, evaluation, and secure API-key handling make the implementation suitable as an academic and research-oriented project.

SAMPLE IMAGES



SmartHire AI

AI-powered career assistant

Features

Resume Analysis

Semantic Job Matching

CV Improvement

AI Career Mentor

Technology

Python

LangChain

FAISS

Ollama

Llama 3.2

Streamlit

Resume Analysis

Extract skills, education, experience and target role from your resume.

Smart Job Matching

Find relevant jobs using semantic similarity and FAISS.

CV Improvement

Identify missing skills and improve your resume content.

AI Career Mentor

Get career guidance using RAG and your career knowledge base.

Analyze Your Resume

Upload your resume in PDF or DOCX format.

Choose your resume

resume4.pdf

2.2KB

+

Selected: resume4.pdf

Analyze Resume

Resume analysis completed successfully!

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Job Matches

Recommended Job Opportunities

Jobs ranked according to the semantic similarity between your profile and the job descriptions.

1. Machine Learning Engineer

Match

78.7%

Job ID: 4

View Job Details

Required Skills

Python, Machine Learning, TensorFlow, Docker, Teamwork, Agile

Description

Develop, train, evaluate, deploy, and maintain machine learning models and pipelines. This role involves working on practical machine learning engineer projects in a collaborative environment.

2. Machine Learning Engineer

Match

78.7%

Job ID: 14

Features

- Resume Analysis
- Semantic Job Matching
- CV Improvement
- AI Career Mentor

Technology

- Python
- LangChain
- FAISS
- Ollama
- Llama 3.2
- Streamlit

AI CV Improvement

Improve your resume based on the strongest matching job.

Personalized recommendations are ready.

Recommendations

1. Missing Skills

The candidate's resume is missing the following skills required by the target job:

- Docker (although mentioned, it's not explicitly stated as a skill)
- Teamwork and Agile (not mentioned at all)

To address this, the candidate can add these skills to their resume, either by adding them to their skills section or by including them in their experience section.

2. Weak Resume Points

The following parts of the candidate's resume are unclear, generic, or weak:

- The "CAREER OBJECTIVE" section is too generic and doesn't provide specific information about the candidate's goals or interests.
- The "EDUCATION" section only mentions the degree and institution, but not the specific courses or specializations that the candidate has.
- The "PROJECTS" section is brief and doesn't provide much detail about the project, such as the technologies used, the problems solved, or the outcomes achieved.

3. Improved Resume Bullets

Here are improved versions of the weak experience and project bullets:

- AI Intern | Innovate AI Labs:
 - Implemented machine learning models using Python and Scikit-learn, and evaluated their performance using metrics such as accuracy and precision.
 - Worked on preprocessing, training, testing, and model evaluation for a machine learning project, ensuring data quality and model reliability.

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Deploy

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- Image Classification System:
 - Developed an image classification application using computer vision and machine learning techniques, achieving an accuracy of 95% on a test dataset.
 - Utilized OpenCV and TensorFlow to build and train a deep learning model, and deployed it using Docker.

4. Improved Professional Summary

Here is a stronger professional summary suitable for the target role:

"Highly motivated machine learning enthusiast with a strong foundation in Python, Machine Learning, and Deep Learning. Proven experience in building and deploying AI-based applications, with a focus on data preprocessing, model training, and evaluation. Skilled in using popular libraries such as Scikit-learn, TensorFlow, and OpenCV, with experience in Docker and collaborative environments."

5. Overall Recommendations

Here are 3 to 5 actionable recommendations for the candidate:

- Add missing skills:** Update the skills section to include Docker and Teamwork/Agile.
- Clarify education:** Add specific courses or specializations to the education section to provide more context.
- Enhance project details:** Expand on the project descriptions to provide more detail about the technologies used, problems solved, and outcomes achieved.
- Tailor the professional summary:** Update the professional summary to focus on the candidate's relevant skills and experience, and to highlight their motivation and enthusiasm for the target role.
- Quantify achievements:** Where possible, use specific numbers and metrics to quantify the candidate's achievements, such as accuracy and precision.



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Deploy

Resume Analysis Job Matches CV Improvement AI Career Mentor

SmartHire AI Career Mentor

Ask questions about careers, skills, jobs, resumes, interviews and professional development.

Suggested Questions

What skills should I learn for Data Science?

How can I improve my resume?

What should I prepare for interviews?

what are the most important question in interviews?

Based on the retrieved context, I don't know based on the available career information. The context only provides information about job responsibilities and job descriptions, but it does not specifically address common interview questions.

Ask SmartHire about your career...



Deploy

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What skills should I learn for Data Science?

Based on the retrieved context, the skills you should learn for Data Science are:

- Python
- Statistics
- Machine Learning
- Data Analysis
- SQL
- Data Visualization

These skills are mentioned as the core skills for a Data Scientist in SOURCE 1: data_scientist.txt.

SmartHire AI • Resume Analysis • Semantic Job Matching • CV Improvement • RAG Career Mentor